

Jeffrey Oduman

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Incoming @ Duolingo June 2026 – Aug 2026

EDUCATION

University of Pennsylvania, Philadelphia, PA

Master of Science in Engineering; Robotics | Accelerated Program

Coursework: Applied Machine Learning, Machine Perception

GPA: Pending

Expected Graduation: May 2028

University of Pennsylvania, Philadelphia, PA

Bachelor of Science in Engineering; Computer Science | AI Concentration

Coursework: Discrete Mathematics, Programming Languages and Techniques, Multivariable Calculus, Social Networks, Probability, Algorithms, Data Structures, Automata, Complexity, Computability, Big Data Analytics, Linear Algebra, Computer Systems, Scalable and Cloud Computing, Operating Systems Implementation & Design, Computer Organization & Design

GPA: 3.78 / 4.00

Expected Graduation: May 2028

SKILLS

Technical: Python, Kotlin, Java, C++, OCaml, SQL, HTML, CSS, JavaScript, Microsoft Office Suite, Google Office Suite, Visual Studio Code, Android Studio, IntelliJ, Git, Tableau, Room, Docker, Confluence, Windows Subsystem for Linux, Ubuntu, Raspberry Pi 4, Arduino, Firebase, Docker, Redis, Java Swing, Node.js, React.js, Pandas, Polars, PyTorch, Matplotlib, JSoup, Jetpack Compose, Robotics-Toolbox, Numpy, Spatial-Math

DL & ML Models used: BERT, FinBERT, Llama.cpp, Faster-Whisper, Random Forest, Logistic Regression, Linear Regression, Gradient Boosting

PROFESSIONAL EXPERIENCES

SafeLab | JoyNet

Philadelphia, PA

Undergraduate Researcher | Full-stack Web Developer | joynet-99b18.web.app

January 2025 - Present

- Built a social media ingestion pipeline drawing ~65,000 TikTok, 20,000 Instagram, and 3,000 Threads posts daily from the EnsembleData API, and processing them through 7 microservices orchestrated via GCP Pub/Sub and Cloud Scheduler, ensuring timely and large-scale data collection.
- Deployed Faster-Whisper for transcription and BERT for sentiment analysis on Google Cloud Run to filter toxic content with 99% accuracy.
- Implemented a social network with user profiles, friends, private messaging, and group messaging, using Firebase Firestore for real-time data and Cloud Storage for media, supporting ~50 daily active users in sharing content, messaging and accessing content with the highest engagement.

Sung Robotics Group

Philadelphia, PA

Undergraduate Researcher | Research award recipient | Compliant Origami Robots

April 2025 - Present

- Developed a gradient descent-based inverse kinematics solver for 3D Dubins Paths, achieving ~28× faster execution (0.5s → 0.018s) compared to particle swarm optimization, while minimizing path length and maintaining axis alignment.
- Optimised Gradient Descent via momentum techniques, adaptive learning rates, and rigorous code profiling, achieving ~3× faster convergence
- Extended 3D Dubins Path theory to robotic arms with 6-DOF and 10-ETS, contributing to newly developing theory in origami linkage structures.
- Built modular tools for visual debugging of 3D paths, reducing validation time by ~50% and enabling adoption by PhD researchers in the lab.

University of Pennsylvania

Philadelphia, PA

Teaching Assistant | CIS 1200 (Programming Languages and Techniques)

January 2025 – December 2025

- Improved instructions and starter code, and added 18+ OCaml/Java test cases for 9 programming assignments, which reduced recurring student errors and resulted in smoother assignment completion for 200+ students, and which professors noted decreased student traffic on ED
- Held 30+ hours of office hours and led 15 weekly recitations; provided individualized debugging support in OCaml/Java and reinforcing problem approach; received consistent positive feedback from students for clear explanations and patient problem-solving guidance.

Cultural.IP

Philadelphia, PA

Software Developer Intern | Intellectual Property

September 2025 – October 2025

- Developed and deployed the Companion platform, a responsive web application enabling startups to manage and monetize intellectual property efficiently across desktop and mobile platforms.
- Spearheaded full migration of the codebase from vanilla HTML, CSS, and JavaScript to a modular React architecture, improving maintainability and enabling component reuse.
- Built the IP Marketplace and Terms of Service Analyzer tools, leveraging JavaScript and third-party libraries to provide real-time, client-facing IP insights.
- Led the closed beta testing program with over 20 early users, collecting feedback and iteratively refining UX, features, and performance metrics.
- Delivered clean, well-documented code in weekly product sprint reviews, and collaborated directly with the CEO on design, implementation, and roadmap priorities.

Penn Electric Racing

Philadelphia, PA

Electrical Software Engineer | Vehicle systems

September 2024 – August 2025

- Co-created [PERDA](#), the team's first custom data visualization library, enabling immediate visualization and aggregation of about 200,000 rows of aerodynamic test data per test run that previously required days of notebook coding.
- Designed C++ code to integrate about 8–10 MS4525DO digital pitot tube sensors into a single aero rake assembly, enabling high-fidelity pressure measurements streamed via I²C during wind tunnel runs for both real-time visualization and aerodynamic model validation.
- Implemented automated tapered charging for custom battery pack accumulators, using PySerial to programmatically control a precision power supply, eliminating manual supervision (saving ~3 hours weekly) and extended pack lifespan by ~20% through safer thermal regulation.
- Debugged and optimized the I²C communication stack to support sensors across subsystems, including differential pressure (MS4525DO), IMUs (MPU-6050), thermal monitors (TMP117), and current sensors (INA219), improving data reliability for aero and powertrain teams

Google Africa Developer Scholarship

Kampala, Uganda

Associate Android Developer | Full stack Android development

June 2022 – May 2023

- Built 6 production-ready Android apps in Kotlin and Java (note-taking app, memory game, weather app, expense tracker, chat/messaging app, and

- to-do list), developing full-stack and abiding by best practices such as accessibility, responsive UI design, MVVM architecture, and REST API.
- Implemented backend services with Firebase Firestore, and Realtime Database, managing structured datasets of up to thousands of records across apps, and used Firebase Auth + GCP Cloud Functions for secure authentication, in-app notifications, and event-driven features.

PROJECTS

- [Quantum Trading](#): a trading terminal built to autonomously take advantage of market opportunities July 2025 - Present
- Built and deployed a cloud-ready trading terminal using Python, JavaScript, and Alpaca API, integrating live market data ingestion, Redis-backed caching, news scraping with Google News RSS Feed, sentiment analysis with FinBERT, and 5 interactive monitoring dashboards
- [Chatbot Service](#): a cloud-hosted TinyLlama-powered chatbot that answers questions about me July 2025 - Present
- Runs Llama.cpp on a cloud-hosted Docker container to minimize inference costs, with security features like CORS, Helmet, and rate limiting
- [NBA Predictor \(group project\)](#): machine learning models to predict NBA game outcomes April 2025 – May 2025
- Led data prep, model training & hyperparameter tuning for the Logistic Regression model (84.7% accuracy, 0.93 AUC). Also led EDA
- [Wiki Olympics](#): a Wikipedia scraper for querying data about the Olympics using JSoup, and Regex March 2025 – March 2025
- Built a modular scraping pipeline to parse unstructured Wikipedias HTML into structured formats using Regex, JSoup and data models

HONORS & AWARDS

Penn Undergraduate Research Mentorship Grant Recipient (2025), Best in South Africa – A Level Computer Science (2025), Best in South Africa – A Level Mathematics (2024), Uganda National High School Chess Gold Medal (2022)